Foundations of Generative AI (ICT4403)

Assignment 3 - Machine Translation with Encoder-Decoder and Decoder-Only Generative Models

Student name	Nehal Verma
Registration number	230953144
Course	Foundations of Generative AI (ICT4403)
Assignment title	Assignment 3 - Machine Translation with Encoder-Decoder and Decoder-Only Generative Models
Title of the developed system	PolyGlot-4: A Comparative English-to-Hindi Machine Translation System across Encoder-Decoder and Decoder-Only Architectures
Dataset name	OPUS-100 (English-Hindi configuration) - Hugging Face id: Helsinki-NLP/opus-100, config en-hi
Dataset link	https://huggingface.co/datasets/Helsinki-NLP/opus-100
Original corpus	https://opus.nlpl.eu/
GitHub repository link	https://github.com/Nehalll-code/fgai-assignment3-mt-comparison
Best-performing model	Gemma-2-2B (LoRA) - BLEU 20.31, chrF++ 38.4 on 500 held-out test sentences

Models implemented

- **Basic Transformer (encoder-decoder):** Trained from scratch; 3+3 layers, d_model 256, joint byte-level BPE (16k)
- **mT5-small (encoder-decoder):** google/mt5-small, fully fine-tuned with a translation task prefix
- **Llama-3.2-1B-Instruct (decoder-only):** LoRA fine-tuned (r=16), plus a 3-shot prompted contrast
- **Gemma-2-2B-IT (decoder-only):** LoRA fine-tuned (r=16), plus a 3-shot prompted contrast

Contents of the repository

- Implementation as Python modules (src/) and a Colab-ready Jupyter notebook (notebooks/).
- README.md with the problem statement, dataset description and link, model descriptions, execution steps and dependencies, results, model comparison and evaluation, sampling-temperature analysis, and the conclusion identifying the best-performing model.
- Evaluation results in results/ - BLEU and chrF++ scores, training and inference times, parameter counts, failure-mode profiling, plots and the generated translations for the full test set.

- docs/responsible_ai.md - the responsible and ethical AI analysis covering bias, hallucination, cultural misrepresentation and sensitive content, with proposed mitigations.